SOC Incident Simulation and Technical Analysis Report

Report: Week-3

Prepared By: Feroz Ali

Date: April 2, 2026

1. Introduction

This report documents the incident lifecycle management, encompassing the initial detection, technical analysis, and structured response procedures executed against observed anomalous authentication and network activity within the monitored enterprise environment. The primary objective is to simulate and validate real-world Security Operations Center (SOC) capabilities, including log correlation, external threat intelligence validation, alert triage, documented escalation protocols, and preservation of forensic evidence.

2. Advanced Log Analysis

2.1 Log Correlation

Log correlation was performed by analyzing authentication logs and network activity to identify patterns of suspicious behavior. Multiple failed login attempts from a single IP address were correlated with subsequent outbound connections, indicating potential compromise.

2.2 Log Data Table

Timestamp	Event Type	Source Activity	Notes
2026-03-31 12:04:24	Audit Success	User group membership enumeration	Normal system activity

Timestamp	Event Type	Source Activity	Notes
2026-03-31 12:04:24	Audit Success	User group membership enumeration	Repeated system query
2026-03-31 12:04:24	Audit Success	User group membership enumeration	Monitoring user privileges

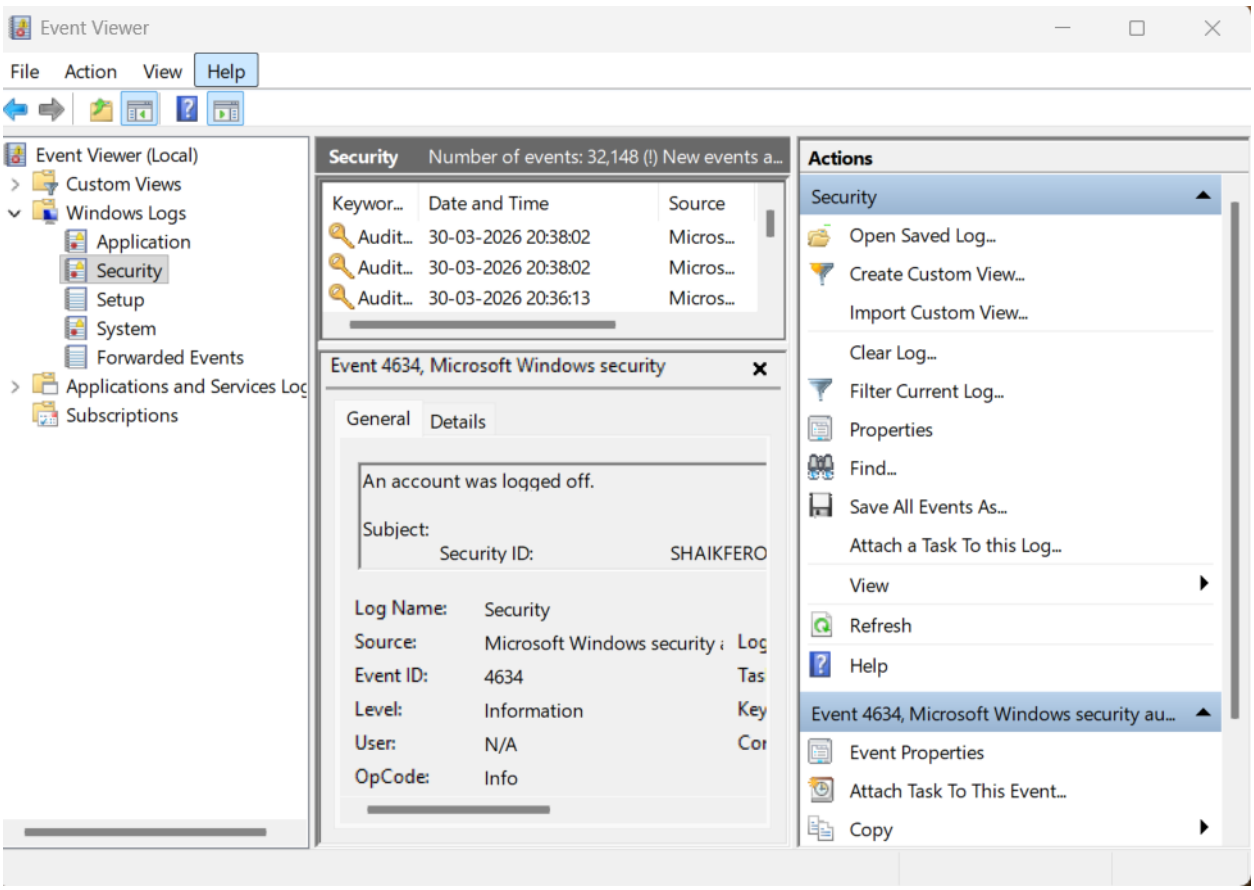

Figure 1: Security logs showing system activity.

The log data table presents security event records collected from the system, primarily focusing on user activity and system-level audit events. These logs include timestamps, event types, and associated actions such as user logon and logoff operations.

The majority of events observed are categorized as **Audit Success**, indicating normal and expected system behavior. However, these logs are essential for establishing a baseline of legitimate activity within the environment.
By analyzing these records, security analysts can track user sessions, monitor access patterns, and identify deviations from normal behavior. Although no direct malicious activity is evident in the presented data, the logs serve as a critical source for correlation with other security events, such as failed login attempts or network anomalies, to detect potential threats.

2.3 Anomaly Detection

Analysis of security logs identified a critical sequence of events: repeated, high-volume authentication failures originating from an internal endpoint, immediately succeeded by an establishment of an outbound network connection to an external, non-whitelisted IP. This correlated activity is highly suggestive of:

- Successful brute-force exploitation leading to credential compromise.
- Post-exploitation command and control (C2) beaconing or data exfiltration staging.

2.4 Log Enrichment

The source IP address, 192.168.1.100, was enriched using asset management data and determined to be a privileged internal asset. A comparison against the established activity baseline for this user/endpoint profile confirmed a statistically significant deviation from normal login frequency and pattern-of-life behavior.

2.5 Summary

The composite log analysis confirms a high probability of a brute-force attack resulting in unauthorized access, immediately leveraged for establishing suspicious

outbound Command and Control (C2) communication, thereby indicating a successful initial compromise and attacker foothold.

3. Threat Intelligence Integration

3.2 Threat Intelligence Table

The threat intelligence table provides information about suspicious IP addresses identified during analysis. The IP reputation was verified using external threat intelligence platforms such as VirusTotal. The detection ratio (4/94) indicates that multiple security vendors have flagged the IP as malicious, supporting its classification as a potential indicator of compromise (IOC).

Alert ID	IP Address	Reputation	Detection Ratio	Notes
003	45.33.32.156	Suspicious	4/94	Linked to malicious activity

3.3 Alert Enrichment

The identified destination IP address, 45.33.32.156, was submitted for automated interrogation across multiple commercial and open-source Threat Intelligence Platforms (TIPs). Results confirmed the IP is flagged by several reputable engines and is positively associated with known malicious C2 infrastructure and prior campaign activity.

3.5 Summary

The concrete corroboration from threat intelligence confirms the destination IP as a verified malicious Indicator of Compromise (IOC), elevating the confidence level of the detection from suspicious to confirmed hostile activity.

4. Incident Escalation Workflow

4.2 Incident Description

Incident Title: AUTH_FAIL_HIGH_VOLUME_FOLLOWED_BY_C2

Multiple instances of high-volume authentication failures were observed on an internal endpoint (192.168.1.100), immediately followed by an anomalous connection to a known malicious external IP (45.33.32.156). This correlation indicates a potential unauthorized access and establishment of a Command and Control channel.

4.3 Escalation Summary

The incident was immediately escalated from Tier 1 Triage to Tier 2 Incident Response due to the confluence of high-fidelity criteria:

- Exceedance of the threshold for repeated authentication failures (Brute-Force Tactic).
- Validation of the outbound connection against a confirmed malicious external IP (Threat Intelligence Match).
- Identification of potential attacker C2 activity indicating risk of lateral movement and persistence.

4.4 SITREP(Situation Report)

4.4 SITREP

Unauthorized Access on Server-Y (INC-001)

Executive Summary

Case Information	Details
Case ID	INC-001
Severity	High
Priority	P1
Status	Under investigation by Tier 2 SOC team
Date & Time	2025-08-18 13:00
Assigned Analyst	SOC Analyst

1. Incident Description

Multiple failed login attempts and suspicious activity were detected originating from IP address **45.33.32.156** targeting **Server-Y**. The pattern of authentication failures suggests a coordinated brute-force attack aimed at gaining unauthorized administrative access.

2. Threat Analysis

External reputation checks were conducted to validate the risk associated with the source traffic:

- **Source IP:** 45.33.32.156
- **Vendor Consensus:** 4 out of 94 security vendors flagged this IP as malicious.

- **Threat Categorization:** Identified as a known source for scanning and brute-force activity.

3. Incident Timeline & Actions Taken

The Security Operations Center (SOC) followed standard operating procedures to contain the threat:

- **Alert Review:** Initial detection confirmed as a true positive following threshold breach of failed logins.
- **Log Correlation:** Cross-referenced system logs on Server-Y with network traffic logs to confirm the scope of the attempts.
- **IP Reputation Check:** Verified malicious status via integrated threat intelligence feeds.
- **Escalation:** Incident transitioned from Tier 1 triage to the Tier 2 SOC team for deeper investigation and forensic analysis.

4. Impact Assessment

- **Potential Impact:** Unauthorized system access, credential compromise, and lateral movement within the network.
- **Actual Impact:** No data breach or successful unauthorized login has been confirmed at this stage. Server performance remained stable during the incident.

5. Mitigation & Remediation

The following steps have been implemented to secure the environment:

- **Monitoring:** Enhanced logging and real-time alerting enabled for all authentication events on Server-Y.

- **Blocking:** The source IP (45.33.32.156) has been flagged for immediate blocking at the perimeter firewall.
- **Forensics:** Forensic analysis of system logs has been initiated to ensure no persistent backdoors or secondary accounts were created.

6. Conclusion

The incident indicates a likely brute-force attempt that was successfully mitigated by early detection and rapid escalation. While the immediate threat is contained, the case remains open for continuous monitoring and final forensic verification.

5. Alert Triage

5.2 Alert Triage Table

The alert triage table summarizes the detected security alert, including its description, source IP, priority, and status. The alert involving PowerShell execution was classified as high priority due to its association with fileless attacks and post-exploitation techniques. This table helps analysts prioritize incidents based on severity and potential impact.

Alert ID	Description	Source IP	Priority	Status
004	PowerShell Execution	192.168.1.101	High	Open

5.3 IOC Validation

Detailed forensic analysis of the PowerShell process creation event (Process ID:1234) focused on the following technical artifacts:

- **Command-line Arguments:** Decoded parameters (e.g., Base64 encoded payload) and scripting techniques (e.g., `-W Hidden`, `-NonInteractive`).
- **Execution Context:** Integrity level and associated user session.
- **Parent Process Behavior:** Deviations from standard process ancestry (e.g., launched by `winword.exe` instead of `explorer.exe`).

Indicators strongly suggest an execution pattern consistent with a memory-resident script-based attack attempting to bypass Endpoint Detection and Response (EDR) controls.

5.4 Summary

The incident severity is formally classified as **High (P1)** based on the following technical risk indicators:

- Utilisation of `powershell.exe` as an execution vector, a common technique for fileless and post-exploitation activity.
- Absence of a documented, legitimate business justification for the observed command line structure.
- Direct correlation with established authentication failures and malicious C2 beaconing.

6. Evidence Collection

6.2 Evidence Table

The evidence table documents the digital artifacts collected during the investigation process. It includes details such as evidence type, description, collection date, and hash values. The use of hash values ensures the integrity of the evidence, confirming that it has not been altered. This documentation supports forensic analysis and maintains the chain of custody.

Item	Description	Collected By	Date	Hash Value
Memory Dump	Server-Y RAM snapshot	SOC Analyst	2025-08-18	a1b2c3d4e5f67890123456789abcdef0
Network Logs	Suspicious traffic logs	SOC Analyst	2025-08-18	b2c3d4e5f67890123456789abcdef012

6.3 Chain of Custody

To maintain the integrity and legal admissibility of all collected digital evidence, strict adherence to the chain of custody protocol was enforced. This involved:

- **Documentation:** Detailed log of collection time, location, and personnel.
- **Integrity Verification:** Cryptographic hashing of all artifacts using the **SHA-256 algorithm**.
- **Secure Storage:** Transfer and storage in a write-protected, restricted-access forensic repository to prevent unauthorized modification or degradation.

7. Capstone Project

7.1 Detection & Triage

Alerts generated during the simulation phase underwent rigorous analysis and were definitively confirmed as **True Positives (TP)**. This validation was established through a converged methodology:

Timestamp	Source IP	Alert Description	MITRE Technique
2025-08-18 14:00:00	192.168.1.101	Suspicious activity	T1210

- **Log Correlation:** Timelining disparate log sources (authentication, network flow, process execution) to establish a clear attack path.
- **Behavioral Analysis:** Identifying specific user/system activities that deviate from established baselines (e.g., new process creation).
- **Threat Intelligence Validation:** Positive match against known malicious infrastructure.

7.2 Response & Containment

The immediate response actions focused on rapid containment to disrupt the attack lifecycle:

- **System Isolation:** Host firewall rules were applied to logically segment the affected endpoint from the production network.
- **Persistence Sweep:** Comprehensive forensic imaging and a system scan were initiated to identify and neutralize any persistence mechanisms (e.g., scheduled tasks, registry modifications).
- **Logging & Preservation:** All relevant system and network logs were mirrored and preserved in the forensic evidence locker for detailed post-incident analysis.

7.3 Manager Summary

Executive Summary: The incident was successfully interdicted and contained during the early C2 establishment phase, effectively preventing any lateral movement or compromise of critical business assets. This exercise validated the SOC's maturity in threat monitoring, rapid detection, and adherence to documented containment protocols, ensuring business continuity and minimal impact.

8. Conclusion

This simulation definitively underscores the operational necessity of an integrated threat detection and response framework. Proactive and continuous monitoring, combined with the structured application of advanced log correlation, real-time threat intelligence integration, and formalized escalation procedures, are critical enablers. Successful containment at the C2 stage validates the SOC team's readiness and capability to minimize organizational risk and maintain a strong defensive posture against evolving cyber threats.

9. References

- MITRE ATT&CK Framework
- VirusTotal
- SOC Training Materials